EECS 16A Foundations of Signals, Systems, & Information Processing
Fall 2025 Discussion 4A

1. Continuous-Time Complex Exponential Potpourri!

Consider the space of continuous-time complex-valued signals over one period, say, over the interval $[0, p]$, where $p > 0$. Define the inner product between two signals x and y as:

$$\langle x, y \rangle = \frac{1}{p} \int_0^p x(t) y^*(t) dt$$

and the induced norm: $\|x\| = \sqrt{\langle x, x \rangle}$.
Let $\phi_k(t) = e^{ik\omega_0 t}$, for $k \in \mathbb{Z}$. Recall that $\omega_0 = \frac{2\pi}{p}$.

(a) **Euler and Periodicity**

(i) Use Euler's formula to express $\phi_1(t) = e^{i\omega_0 t}$ in terms of sine and cosine.
Answer: By Euler's formula: $e^{i\omega_0 t} = \cos(\omega_0 t) + i \sin(\omega_0 t)$.

(ii) Show that $\phi_k(t)$ is periodic with fundamental period p for any integer k.
Answer: We check $\phi_k(t+p) = e^{ik\omega_0(t+p)} = e^{ik\omega_0 t} e^{ik\omega_0 p}$. Since $\omega_0 p = 2\pi$, we have $e^{ik\cdot 2\pi} = 1$, so $\phi_k(t+p) = \phi_k(t)$. Thus, $\phi_k(t)$ is periodic with period p. Since k is an integer and the minimal multiple T satisfying $k\omega_0 T = 2\pi m$ is $T = p$ when $m = k$.

(iii) Is $\phi_{1/2}(t) = e^{i(\omega_0/2)t}$ periodic with period p? Justify. If not, what is its fundamental period?
Answer: Check: $\phi_{1/2}(t+p) = e^{i(\omega_0/2)(t+p)} = e^{i(\omega_0/2)t} e^{i\pi} = e^{i(\omega_0/2)t}(-1) \neq \phi_{1/2}(t)$. So not periodic with period p. The fundamental period T satisfies $(\omega_0/2)T = 2\pi \Rightarrow T = \frac{4\pi}{\omega_0} = 2p$.

(b) **Signal Space and Norms**

(i) Find $\|\phi_k\|$ for any integer k.

Answer:

$$\|\phi_k\|^2 = \langle \phi_k, \phi_k \rangle = \frac{1}{p}\int_0^p e^{ik\omega_0 t} e^{-ik\omega_0 t} dt = \frac{1}{p}\int_0^p 1 dt = 1.$$

So $\|\phi_k\| = 1$. All complex exponentials have unit norm, i.e, they are normalized vectors in this space.

(ii) Compute $\langle \phi_k, \phi_m \rangle$ for integers $k \neq m$.

Answer:

$$\langle \phi_k, \phi_m \rangle = \frac{1}{p}\int_0^p e^{ik\omega_0 t} e^{-im\omega_0 t} dt = \frac{1}{p}\int_0^p e^{i(k-m)\omega_0 t} dt.$$

For $k \neq m$, this is:

$$\frac{1}{p}\left[\frac{e^{i(k-m)\omega_0 t}}{i(k-m)\omega_0}\right]_0^p = \frac{1}{i(k-m)\omega_0 p}\left(e^{i(k-m)2\pi} - 1\right) = 0,$$

Since $e^{i2\pi n} = 1$ for integer $n = k - m$, we have:

$$\frac{1}{i(k-m)\omega_0 p}(1-1) = 0.$$

So inner product is 0.

(iii) Based on (i) and (ii), argue why the set $\{\phi_k(t)\}_{k\in\mathbb{Z}}$ forms an orthonormal set over $[0, p]$.

Answer: From (i), $\|\phi_k\| = 1$ for all k (normal). From (ii), $\langle \phi_k, \phi_m \rangle = 0$ for $k \neq m$ (orthogonal). Together, this means the set is orthonormal.

(c) **Linear Independence and Span**

Consider the finite subset $\mathscr{B} = \{\phi_{-1}(t), \phi_0(t), \phi_1(t)\}$.

(i) Show that the signals in $\mathscr{B}$ are linearly independent over $\mathbb{C}$.

Answer: To prove linear independence, we are interested in a linear combination of these signals that equals zero:

$$c_{-1}\phi_{-1}(t)+c_0\phi_0(t)+c_1\phi_1(t)=0,$$

where $c_{-1},c_0,c_1 \in \mathbb{C}$.
Recall from previous parts that the inner products $\langle \phi_k(t),\phi_l(t)\rangle$ for $k,l \in \{-1,0,1\}$:
1. For $k=l$:

$$\langle \phi_k(t),\phi_k(t)\rangle = 1.$$

2. For $k \neq l$:

$$\langle \phi_k(t),\phi_l(t)\rangle = 0.$$

Now, we take the inner product of the equation with $\phi_{-1}(t)$:

$$\langle \phi_{-1}(t), c_{-1}\phi_{-1}(t)+c_0\phi_0(t)+c_1\phi_1(t)\rangle = \langle \phi_{-1}(t),0\rangle = 0$$

Expand:

$$c_{-1}\langle \phi_{-1}(t),\phi_{-1}(t)\rangle + c_0\langle \phi_{-1}(t),\phi_0(t)\rangle + c_1\langle \phi_{-1}(t),\phi_1(t)\rangle = 0$$

Simplify with orthonormality:

$$c_{-1}=0$$

Doing the same with $\phi_0(t)$ and $\phi_1(t)$ we get $c_{-1}=c_0=c_1=0$. So the signals $\phi_{-1}(t),\phi_0(t),\phi_1(t)$ are linearly independent over $\mathbb{C}$.

(ii) Let $x(t)=3+2\cos(\omega_0 t)+4\sin(\omega_0 t)$. Express $x(t)$ in the $\mathscr{B}$ basis.
Answer: Use inverse Euler's: $\cos(\omega_0 t)=\frac{e^{i\omega_0 t}+e^{-i\omega_0 t}}{2}$, $\sin(\omega_0 t)=\frac{e^{i\omega_0 t}-e^{-i\omega_0 t}}{2i}$. Then:

$$x(t)=3\cdot e^{i0\omega_0 t}+2\cdot\frac{e^{i\omega_0 t}+e^{-i\omega_0 t}}{2}+4\cdot\frac{e^{i\omega_0 t}-e^{-i\omega_0 t}}{2i}$$

$$x(t)=3\cdot\phi_0(t)+2\cdot\frac{\phi_1(t)+\phi_{-1}(t)}{2}+4\cdot\frac{\phi_1(t)-\phi_{-1}(t)}{2i}$$

Simplify:

$$x(t)=3\phi_0(t)+(1-2i)\,\phi_1(t)+(1+2i)\,\phi_{-1}(t).$$

(iii) What is the dimension of span($\mathscr{B}$)? Describe the class of signals that live in this span.

Answer: Since $\mathscr{B}$ has 3 linearly independent vectors, $\dim(\text{span}(\mathscr{B})) = 3$. The span consists of all signals of the form $ae^{-i\omega_0 t} + b + ce^{i\omega_0 t}$ with $a, b, c \in \mathbb{C}$, which correspond to real-valued sinusoids up to frequency ω_0 (i.e., DC + first harmonic). Nice!

